Biomedical Semiconductor Devices

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Abstract:

Biomedical semiconductor devices stand as one of the paramount technological advancements for healthcare and humanity as a whole, providing new solutions for diagnosis, treatment, and monitoring of individuals. This paper will introduce the significance of technology in healthcare and elucidate the concept of biomedical semiconductor devices. Following this, the fundamentals, diverse applications, challenges and limitations of these devices will be covered, diving deeper into the unimaginable capabilities of these devices and crucial concerns to consider such as their reliability and biocompatibility. Given the importance of these devices, their optimization and advancement is an industry in which hundreds of billions of dollars is dedicated yearly. Therefore, this paper delves into the recent advancements and future prospects of these devices, highlighting how these devices become more and more integrated and useful to people's everyday lives. Finally, conclusions will be made regarding biomedical semiconductor devices and the significance of the research/advancements made in this industry.

Introduction:

As human society continues to make rapid technological advancements, the convergence of these semiconductor technologies with the biomedical engineering industry has led to incredible developments in healthcare. In today's world, semiconductor technology is of utmost necessity in every facet of healthcare, from daily checkups with blood pressure monitors and electronic thermometers to crucial devices such as medical imaging equipment and implantable cardiac devices.

Semiconductors are a component of electronic devices which are made from pure "semiconductor" elements, such as silicon, germanium, or gallium arsenide as common examples. These elements are considered semiconductor materials because when they are methodically exposed to small amounts of impurities, known as doping, the conductivity of these elements changes drastically, allowing them to behave both as a conductor and as an insulator when desired. These components are so useful when it comes to biomedical applications as many of these materials are biocompatible, and given their electrical properties and sensitivity to stimuli, semiconductors can be applied to essentially all aspects of human health. This is what makes these devices so valuable in the healthcare/biomedical industry, confirming why they

generate hundreds of billions of dollars invested yearly. Imagine life today if you couldn't go to the doctor and be properly diagnosed with injury/disease/sickness through medical imaging technologies or blood analyzers such as hematology or chemistry analyzers? Imagine if you required a major invasive surgery, and technologies/devices used on you were still primitive? While these were realities at one point in time, nowadays these realities are unimaginable. It is for this reason that this area of study is paramount, as continued exploration and understanding of these technologies are essential for nurturing the next generation of engineers and advancing further innovations in healthcare.

Fundamentals of Biomedical Semiconductor Devices:

Regarding understanding biomedical semiconductor devices, it is important to first address what exactly these semiconductor materials are at a molecular level, allowing these devices to exist. Materially, "Silicon (Si) has been the material of choice for numerous wearable and implantable [biomedical] devices because of its low cost, global availability, and standard industrial manufacturing practices" [1]. Being able to create flexible Si devices allows for applications such as soft wireless-communication modules and cardiac electrophysiology recording devices which allow the rigidness of the device to shape itself with the human tissue it lays upon. In this way, its high electron mobility and low intrinsic carrier concentration allows Si to have great precision over electrical signals, and its biocompatibility and compatibility with standard fabrication processes allows for cheap and mass production and flexibility to integrate effectively with biological tissue. Although this may be the case, there are many drawbacks to the use of Si, "such as fast ion diffusion and degradation in bio-fluids, making them unsuitable for numerous wearable and implantable applications, especially for long-term implanted devices" [1]. For this reason, engineers began to venture from Si into a different area of semiconductor materials, known as Wide-Band-Gap (WBG) materials.

WBG materials are "defined as materials possessing a band gap of 2.2 electron volts (eV) or higher" [1], which allows this specific family of semiconductor materials to have a high breakdown electric field and emit short wavelengths. Due to this, many WBG materials such as Zinc Oxide (ZnO), Gallium Nitride (GaN), and Silicon Carbide (SiC) are able to be used in biofluidic environments and have long-term stability, with potential uses being listed in Figure 1. This has led to these WBG materials holding a very large part in today's semiconductor market,

due to their superior properties to Si for biomedical devices, allowing development of devices such as laser diodes, photodetectors, and microfluidic devices.

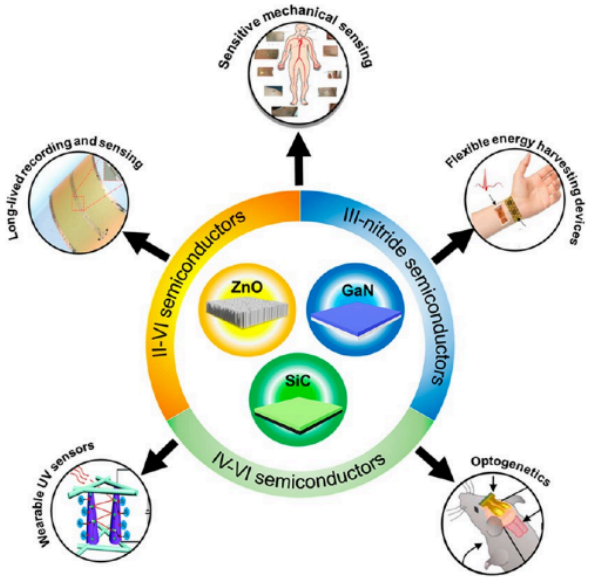

Figure 1: Commonly used wide-band-gap materials and their applications. [1]

While there are numerous materials that are conducive to the design of biomedical semiconductor devices, there are also numerous fabrication techniques tailored for their development. The first type of fabrication technique used in the general substrate design process is patterning techniques such as photolithography, electron beam lithography (EBL), and ion beam lithography (IBL), techniques that "have enabled fabrication of nano-channels with less than 100nm dimensions" [2]. What these techniques essentially do is define patterns on semiconductor substrates, either through UV light or a focused beam of electrons or ions. Depending on resolution requirements and cost requirements, engineers may choose to employ methods such as photolithography and UV light for a device which requires less precision and has a lower project budget, and may choose to use IBL and a focused ion beam for a substrate that requires very precise patterns and a large budget. With these patterning processes, the substrate is able to have the layout of its transistors, electrodes, interconnects, etc. physically designed, allowing engineers to specifically design these devices to their biomedical applications.

After the patterning techniques, the next crucial fabrication techniques are the deposition techniques, specifically physical and chemical deposition. Typically, physical deposition is done through processes such as tilted e-beam evaporation, spin coating, or sputtering, where physical

processes are performed to deposit/transfer atoms onto the substrate and form a thin film. In tilted e-beam evaporation, a focused electron beam is used to heat a material and vaporize it. The vaporized material then condenses on the semiconductor substrate, forming the desired film layer and completing the deposition. On the other hand, processes such as sputtering deposition as can be seen in Figure 2, are when a material is bombarded with energetic ions, causing atoms to eject from the material's surface and onto the pre patterned trenches. Often hand in hand with physical deposition methods come chemical ones such as chemical vapor deposition (CVD), which is similar to tilted e-beam evaporation but utilizes chemical reactions between elements in their vapor form in order to create new layers on a substrate. Therefore, if for example a substrate was spin coated with thin film layers, these layers could be used as precursor coatings for a CVD to create the desired final layer on the substrate. Common thin layer materials such as silicon dioxide (SiO2) can be made by using silicon and oxygen precursor material in the sputtering or spin coating deposition and then completing the layer with CVD.

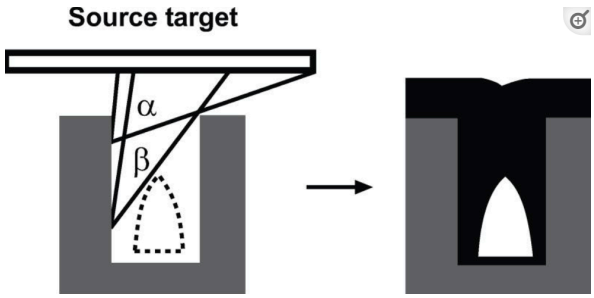

Figure 2: A schematic illustration of the sputtering deposition process to form sealed nano channels on pre-patterned nano-trenches utilizing structural shadow effects.

After deposition, the next fabrication technique which is crucial in the development of biomedical semiconductors is etching/doping. While the patterning and deposition techniques are used as a mask, etching and doping is where the actual addition and removal of material takes place to create the final semiconductor substrate patterns. Etching, which is the process of removing material from the substrate to form the desired transistor and component patterns, is a very common and crucial fabrication step which refines the patterns and layers developed in the pattern and deposition stages. The significant difference in a substrate's pattern before and after etching can be seen in Figure 3, highlighting the importance of this step in developing the final product. Unlike etching, doping introduces impurities to the substrate, physically modifying the electrical properties of the substrate. In doing so, engineers are able to create different p-type and n-type conductivity regions, changing the carrier concentration in specific regions of the

substrate and therefore changing how the transistors and components interact with each other on the semiconductor interface. This allows engineers to essentially control the functionality of the final biomedical device through the doping of its components.

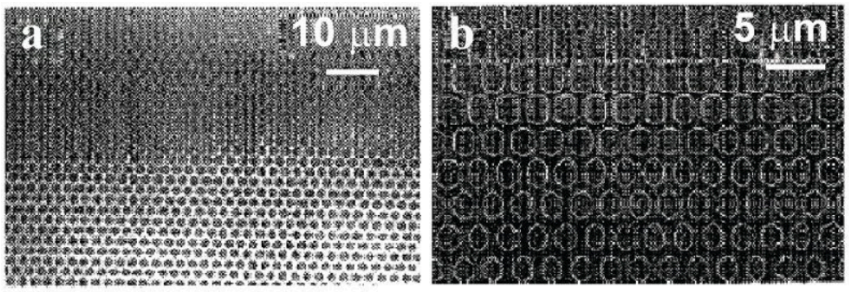

Figure 3: **a**) Optical image of the sample after photoresist development (photolithography). **b**) Optical image of the sample after pattern transfer and photoresist removal (etching). [2]

One final technique used in the fabrication process that is quite often overlooked are the packaging techniques. This step is vital because once a semiconductor product is manufactured, it is essential to uphold the reliability, functionality, and overall integrity of the device and its components during shipping and usage. This is typically done through encapsulation processes such as thin–film encapsulation or wafer bonding. Thin-film encapsulation "requires deposition of various thin films […] forming a thin but hermetically sealed capping layer [...] provid[ing] a low-cost packaging solution that eliminates the need for a separate cap wafer and bonding equipment" [3]. Although a lower cost method, due to the capping layer being made of thin films, this type of encapsulation typically requires additional mechanical protection. On the other hand, wafer bonding "employs a separate cap wafer for the sealing purposes, [providing] perfect mechanical robustness for the protection of encapsulated MEMS sensors, also [allowing] very low package pressures" [3]. In this way, there are also encapsulation methods that are more protective although potentially more costly, illustrating that a lot of the design process falls under the engineers and deciding what is best for the project at hand. For a biomedical product with a low budget and low protection priority such as an electronic thermometer for example, economically it may not be reasonable or justified for a team of engineers to use more expensive protection methods on such a cheap low budget product. On the other hand, products such as bioMEMS devices which utilize a lot of expensive micro components and MEMs sensors, may require engineers to use more expensive and rigorous protection methods in order to ensure the integrity and functionality of the devices and prevent damage from environmental/external

factors. An example of wafer bonding for a MEMS sensor on a glass substrate can be seen below in Figure 4.

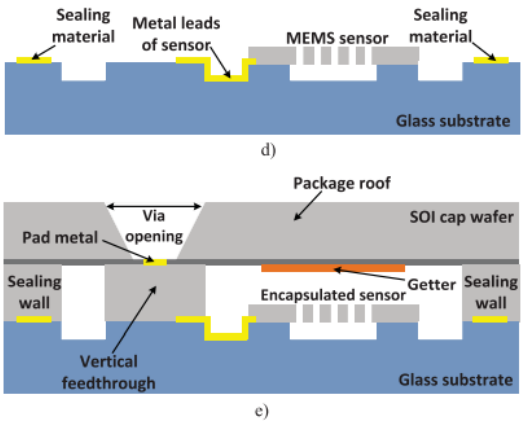

Figure 4: d) Fabrication of MEMS sensor wafer using M-SOG technology. e) Wafer level hermetic encapsulation (Wafer Bonding) with Au-Si eutectic bonding. [3]

In this way, there are a multitude of materials and fabrication techniques that are generally used in the creation of biomedical semiconductor devices, illustrating not only the advanced nature of these technologies but the endless number of ways these semiconductor substrates can be designed to solve both current and future biomedical problems.

Applications of Biomedical Semiconductor Devices:

In today's world, these semiconductors are the backbone for a myriad of groundbreaking biomedical applications, facilitating the realization of some of the world's most awe-inspiring technological feats. One technological sector in which biomedical semiconductors have revolutionized is that of imaging technologies. X-rays, which were discovered "by W.C. Röntgen in 1895, [established X-ray imaging] as the first medical imaging technique"[4]. From then on, medical imaging technologies only became more and more advanced, with processes such as positron emission tomography (PET) and magnetic resonance imaging (MRI) also paving their way to the forefront of imaging technology. X-ray technology itself has also become much more advanced and has branched out in many different ways thanks to the continuous development of X-ray detectors, built upon semiconductor components. With these detectors, surgeons and doctors are able to perform imaging techniques such as radiography, mammography, and computed tomography (CT) scans. Radiography, "which covers chest, trauma, pediatric, orthopedic and general examinations has evolved [...] from a technique dominated by analog

screen-film combinations to a digital technique based on flat detectors" [4]. In doing so, flat detectors were able to develop the modern digital X-ray technology that we use today in all X-ray practices, leading to the development of complex technologies such as robotic c-arm systems for radiology, cardiology, oncology, or surgery environments [4] as seen in Figure 5. Thanks to these detectors' ability to operate with hydrogenated amorphous silicon and thin-film transistors, X-ray detectors have become the model for modern day imaging technologies.

Figure 5: Robotic C-arm system used in a hybrid operating room for endovascular aortic repair including coiling of the arteria mesenterica inferior.[4]

Furthermore, PET scanning technology frequently accompanies that of CT scans. Positron emission tomography entails administering a radioactive tracer into the body, releasing positrons, and subsequently converting the "scintillation light produced by gamma-ray interactions in the crystal into an electronic signal" [5]. In doing so, information can be collected regarding an individual's physiological processes, enabling diagnoses to be made regarding conditions such as cancer, cardiovascular diseases, and neurological disorders. For this reason, pairing PET with CT proves to be very useful, as it can use one hybrid PET/CT machine that creates hybrid images which capture both metabolic and structural data about an individual simultaneously, allowing for more accurate findings. In fact, "in about a 4-year period, dedicated PET went from 100%-0% market share, being completely replaced by hybrid PET/CT in new systems that [were] sold" [5], illustrating the dominance of combining these technologies in diagnostic care over traditional PET methods.

Similarly, PET is also often paired with another form of advanced imaging known as magnetic resonance imaging. MRI, a "technique based on the nuclear magnetic resonance (NMR) phenomenon" [6], allows doctors to obtain very detailed and high-contrast images of an

individual's internal structures. MRI is a different approach than that of CT and PET for example as the "technique uses non ionizing radiation, relying on static and slowly changing magnetic fields and electromagnetic (EM) energy [...] provid[ing] soft- and hard-tissue images with outstanding contrast that can be both static and dynamic" [6]. This proves the combination of PET and MRI highly beneficial, as it enables the simultaneous capture of an individual's physiological processes and anatomical structures like soft tissues and organs. This capability facilitates extremely detailed imaging and consequently enhances diagnostics, as depicted in Figure 6. In this way, modern imaging technologies significantly contribute to the biomedical industry by facilitating early detection, accurate diagnosis, and effective treatment of medical conditions, allowing for a happier and healthier society.

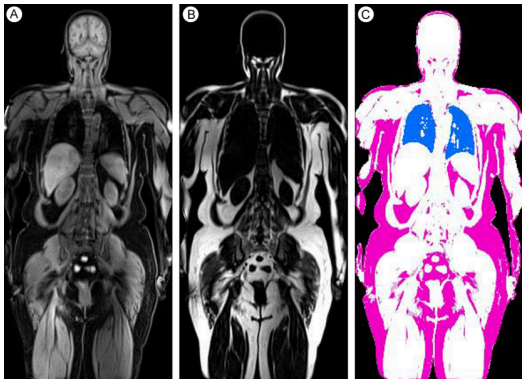

Figure 6: MR-AC data generated by the 2-point Dixon method. From left to right: (A) water image, (B) fat image, and (C) segmented 4-class attenuation map (lean tissue ¼ white, adipose tissue ¼ pink, lung ¼ blue, air ¼ black).[5]

Another technological sector within the biomedical semiconductor device field that holds a significant market share is that of biosensors, specifically those for the monitoring/sensing of DNA, glucose, and proteins. Biosensors, which are devices made to detect and/or quantify specific biological molecules or analytes, utilize many different types of semiconductor sensors in order to achieve these goals, "includ[ing] optical, electrochemical, electrical, gravimetric, acoustic, and magnetic types" [7]. In doing so, these biosensor devices are able to be customized by engineers in order to detect specific nucleic acids and proteins with high detection sensitivity. At present, fluorescence-based sensors are regarded as the preferred choice in the industry for both homogeneous and heterogeneous DNA and protein measurements, owing to their remarkably high detection sensitivity [7]. But, there have also been many other types of sensors emerging to the forefront, such as fiber optic and Forster resonance energy transfer (FRET) sensors, being able to detect nucleic acid and protein binding events at the individual molecular

level. Essentially, this device market is such a massive and significant one in the biomedical industry due to this technology's ability to detect one thing; biomarkers. What makes the detection of biomarkers in humans so important is that “they provide early warnings of disease and are active monitors that translate patient responses to treatments” [7].

A highly regarded and researched biomarker is glucose, given its significance in diagnosing and treating diabetes, a condition affecting hundreds of millions globally. Glucose sensors typically come in two different forms: enzymatic and non-enzymatic glucose sensors. Enzymatic glucose sensors employ semiconductor oxide electrodes to detect oxygen consumption facilitated by glucose oxidase, while non-enzymatic glucose sensors directly oxidize glucose in the sample [7]. This demonstrates that even within a single type of sensor, there are various methods to tailor the sensor for specific requirements. For instance, enzymatic glucose sensors employ transduction materials like FETs, optical sensors, and electrochemical transducers. On the other hand, non-enzymatic glucose sensors, which are less common in the industry, use materials such as metal oxides, carbon and nano materials. Therefore, for example, if engineers are working on a continuous glucose monitor for individuals at home with diabetes, a device like this would need to be very accurate and sensitive as it could be the difference between life and death. For this reason, they would most likely construct a high-end enzymatic glucose monitor, using enzymatic glucose sensors made of FETs, electrochemical transducers, and/or optical transducers. If another group of engineers were working on a low-budget environmental project detecting glucose in water samples, they may choose to use non-enzymatic sensors as they are more cost-effective, and perfect accuracy/sensitivity is not as critical. Hence, biosensors emerge as a crucial technological advancement in the biomedical sector both presently and in the foreseeable future, with their semiconductor technologies offering a broad spectrum of medical benefits, spanning from enhanced research capabilities to lifesaving interventions.

One final major application sector of biomedical semiconductors is that of therapeutic devices such as neural implants and drug delivery systems. Invasive and implantable neural implants, which are quite a recent and major technological feat in the biomedical industry, could come to be thanks to advances in the imaging devices discussed earlier such as MRI/PET, allowing for doctors and engineers to better understand how the brain and its neuronal circuits work. While much research is currently being done on the development of soft, flexible

implantable neural devices, most existing chronic neural devices are tethered and rigid devices. To combat their lack of biocompatibility, these semiconductor devices utilize a wireless power system, with "the most commonly used wireless power technologies used [being] electromagnetic, photovoltaic, and ultrasound" [8]. With these types of wireless semiconductor technologies having their own individual properties, each type can be "used to their strengths" in order to address specific biomedical problems. For example, with electromagnetic based power transfer neural devices, because of their use of electromagnetic induction, requires less complex circuitry and provides a larger power transfer distance, making it very compatible with miniature neural devices and therefore very reliable under long-term implantation. For this reason, electromagnetic neural implants tend to be used for a majority of these devices which are chronic implants, such as deep brain stimulators (DBS), spinal cord stimulators (SCS), and responsive neurostimulation systems (RNS), which are "peripheral and spinal nerve interfaces for monitoring epilepsy [and aborting epileptic seizures]"[8]. Because of these devices being used to save peoples' lives, operating directly on the central nervous system, their reliability is paramount, and therefore engineers can use their knowledge of the different semiconductor properties in order to cater to these biomedical objectives. Photovoltaic neural implants, for example, use photovoltaic cells to convert optical signals into electrical impulses, making their properties perfect for objectives such as creating retinal implants for those with retina damage/disease.

This leads into the other major category of therapeutic biomedical semiconductor devices, drug delivery systems, which commonly use wireless power technologies. These drug delivery systems are so important for society because there are many chronic illnesses/diseases where in which traditional drug therapy does not do the patient service, as typical drug administration is typically non-targeted, and administrations are far and few between, meaning that there are major fluctuations of drug levels within the body as opposed to a continuous regulated level. What is great about a drug delivery system is that it almost behaves like a closed-loop system, continuously monitoring the level of drug presence at the target site, continuously reacting to the desired drug levels tailored by the doctor for the individual. An example of an implanted drug delivery system that uses MEMS are silicon micropumps, first made for insulin infusion in diabetic patients [9]. Therefore, rather than just monitoring glucose like the sensors previously, these micropumps are able to integrate semiconductor technology to

flow out liquid insulin into a patient's body when necessary at high precision, "taking action" on the monitored data. For targeted drug delivery release with non-implanted drug delivery devices, the industry tends to lean towards bio-capsules and microparticles, as their micro-sized nature allows them to "bind ligands to the surface of microparticles" [9], allowing them to target delivery to the cellular level. While bio-capsules aren't semiconductors themselves, they are often integrated with semiconductor technology for data collecting purposes, just like that of microneedles. Micro-machined silicon microneedles and/or multi-channeled silicon probes, which follow a similar fabrication process to that of semiconductor devices, are commonly integrated with semiconductor devices such as the sensors discussed before or CMOS circuitry in order to regulate the "transdermal delivery of drugs over an extended period of time" [9]. In doing so, "transdermal drug delivery overcomes the limitation of gastrointestinal drug degradation and eliminates the pain and inconvenience of intravenous and intramuscular injection" [9]. As can be seen in Figure 7, these silicon probes are very integrated and micro-sized, allowing for monitoring and delivery of drugs at a targeted microscopic level.

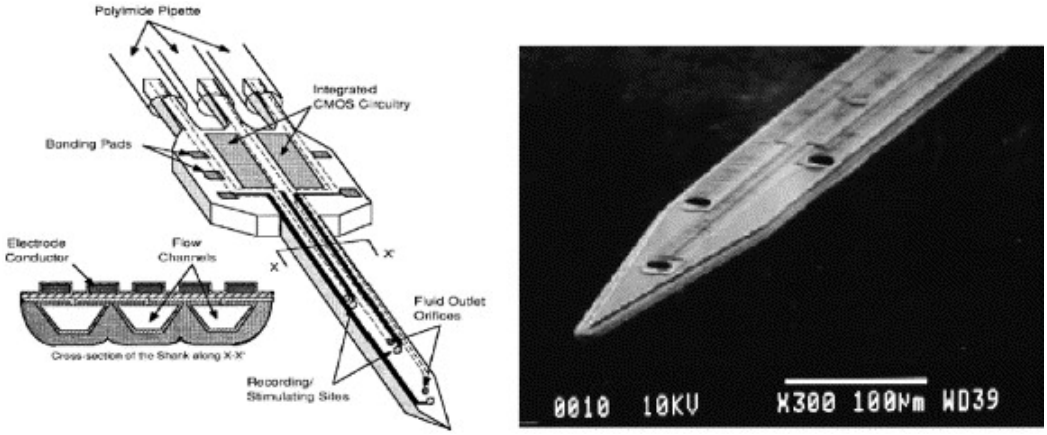

Figure 7: Micro-machined neural tissue drug delivery probe. Left: Sketch of a probe having three delivery channels along with recording and stimulating electrodes. Right: SEM view of a probe with three delivery channels[9].

In this way, drug delivery systems are very much related to and even more crucial to the biomedical industry than biosensors, as these systems work hand in hand with biosensors, neural implants, etc. to collect real-time data on a patient and deliver drugs accordingly. As a matter of fact, all of these biomedical semiconductor devices are very much related to one another, and the knowledge/advancements collected in one device sector only makes it easier to learn more and improve the devices in all the other sectors, as the term goes; "iron sharpens iron". For example, with better imaging tech, engineers are able to learn more about the brain and build better neural

implants, and better neural implants can lead to more effective drug delivery systems, which may lead to more accurate sensors, and so on. On this basis, it is no accident that the biomedical semiconductor industry has already improved the quality of life of billions of people over the past several years, with major feats still waiting on the near horizon.

Challenges and Limitations:

Just like any other project or technology to ever exist, there have been, will be, and currently are a multitude of challenges and limitations when it comes to the development of biomedical semiconductor devices. First and foremost when discussing biomedical devices, the main unique challenges that come to mind are those regarding device biocompatibility. While all biomedical semiconductor devices do not require biocompatibility, many of them, especially the very expensive and important devices, interact directly with human tissue/cells and therefore require significant biocompatibility characteristics. Currently, "the most common materials used to interface between biological tissue and conventional inorganic electronic materials are hydrogels" [10], but the issue with these materials are the fact that they are not semiconductors, therefore, engineers cannot reap all of the bioelectronic benefits that come with the use of semiconductors. The primary challenges associated with semiconductor devices stem from the materials used in their construction, as only a few semiconductor materials are biocompatible. In order to be biocompatible, the tissue response to these devices must be positive, therefore these materials must be non-toxic, non-bioinert, and either bioactive or bioresorbable. There are many different ways that these materials could prove to be toxic to the body, and these are "assessed by tests of cytotoxicity, sensitization, irritation, systemic toxicity (acute), sub-acute and sub-chronic toxicity, genotoxicity, hemocompatibility, chronic toxicity, carcinogenicity, reproductive toxicity, biodegradation, etc." [10]. Therefore, right off the bat, engineers are already set with a limited list of available materials when tasked with building a solution semiconductor device; but this list only gets smaller. This is because in order to compete with the likes of hydrogels, which are very flexible and conductive, the materials selected by engineers need to also prove to be sufficiently conductive and flexible to be compatible with soft and flexible tissue; a difficult ask for the typically mechanically rigid semiconductor device. Additionally, selected semiconductor materials must not be susceptible to biofouling while also still being cell adhesive, therefore

these devices can effectively interact with desired cells while also not interacting with undesired biological materials in the same area.

This goes to show that engineers already face a lot of obstacles in device development just in biocompatibility issues alone, but these aren't the only challenges that engineers face. Beyond the issues of needing high biocompatibility and low toxicity, “general requirements are [...] minimal size and weight, low power consumption, good reliability,[...] high data rate and data latency” [11]. The reason why aspects like low power consumption, reliability, and high data rate are so important for these types of devices is because, especially for implantable devices “with single-use batteries, the replacement of such a device would require a costly and invasive surgical procedure” [11]. Therefore, for single-use battery devices, essentially, the goal is to have it last for a lifetime, that way, surgical intervention would never be required in order to replace the battery. At the same time, the device needs to be able to consume enough power to complete whatever task it was designed to do in the body, and with these two design goals working directly against each other, the challenges stack up against engineers. Something like a pacemaker for example, while its power consumption can be maximized, can’t last in the body for a lifetime as it needs to consume enough power to function, meaning eventually after like 10 years or so, the battery would need to be replaced. A rechargeable power source may seem like a potential answer for this issue, but frequent recharging can be a very resource and time-consuming activity [11]. This is where the concept of energy harvesting comes in, where devices capture and convert the surrounding energy into usable electrical power. For biomedical devices, energy harvesting can come in two different forms as seen in Figure 8, either from the human body itself by means such as thermal or kinetic energy or by means of the surrounding environment. What makes energy harvesting such a groundbreaking solution to durability and reliability concerns is the fact that if a device can continuously provide energy to itself and becoming self-sufficient, barring strong and durable device interconnects in fabrication/development process, a device can theoretically prove to be both durable and reliable for an entire lifetime without intervention. So while engineers are faced with a multitude of challenges with biomedical devices, a multitude of solutions are also being brought to the forefront of the industry.

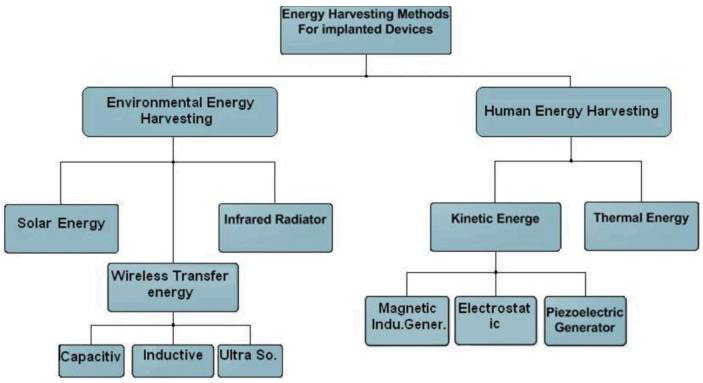

Figure 8: Energy harvesting methods used in biomedical applications. [12]

Finally, just like all other real-world projects, even engineers have to keep business and economic concerns in mind. While all engineers operate under the general concept of "kaizen", or continuous improvement, engineers have to keep this in mind for all aspects of product development, including economic improvement. Therefore, while it is always possible to improve the technology for a given device, engineers are limited by and have to keep in mind the economic goals for a given project. For example, a simple take-home glucose testing kit should be relatively cheap for the end user and shouldn't require extreme precision accuracy. Therefore, businessmen want to keep the production of this product cheap, therefore engineers must keep in mind that something such as a simple glucose biosensor would suffice in the development of this product and its goals. On the other hand, something like a continuous glucose monitor, which someone would wear to get continuous real-time monitoring of their glucose levels, would need the device to be highly reliable and efficient. For this reason, engineers would need to develop a device which utilizes much more advanced technology and biosensors, so the economic scope of the product's purpose also needs to be kept in mind. So while it may seem like kaizen to develop something like a cheap take home glucose testing kit with the same advanced technology as a CGM, the economic improvement of the product is not also being considered. Similarly, engineers may be faced with a ton of regulatory hurdles and/or standards compliances, making what they want to do not necessarily the same as what they are allowed to do. Due to the aspect of these devices being used commercially, a ton of quality control, regulation, and safety testing needs to be done on devices before they are able to actually be used. This can lead to years and years of hold up, and many different prototype trials and redevelopments. This illustrates just how many different parties are interested/involved in the development of biomedical semiconductor devices, highlighting that engineering challenges go beyond math and science.

Recent Advances and Innovations/Future Directions:

While the biomedical semiconductor industry already has made wondrous technological feats, this industry is one which is still growing at a rapid rate, proving to develop many new semiconductor technologies with even more promising futures. For one, engineers are always striving for technological efficiency, and for biomedical semiconductor devices, this usually comes in the form of minimizing device size. For biomedical devices, the presence of the device and the process of insertion of the device are desired to be invisible and simple, and the smaller the devices become, the more achievable this goal becomes. For this reason, nanotechnologies have become the main sector of research for future biomedical device development, whose market revenue/size growth can be seen in Figure 9. The way is being paved for nanoscale sensors which can detect and prevent post-surgery inflammation and infection, as well as nano therapies and nanoparticles which could have the ability to completely reverse type-II diabetes and deliver drugs exclusively to cancer cells as an alternative to chemotherapy [13]. With the growth of nanotechnology development, it is clear how this technology has the capability to revolutionize the future of medical treatment and save the lives of millions.

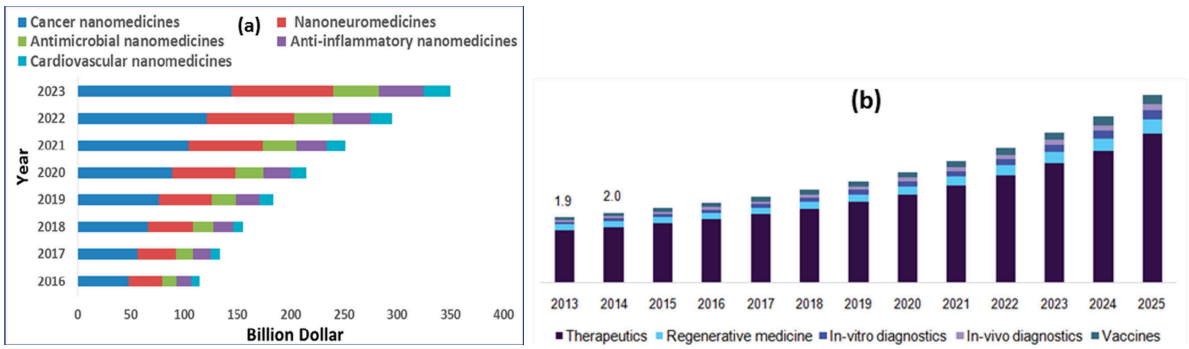

Figure 9: (a) The global nanomedicine market revenue trends and (b) US nanomedicine market size for by-products. [13]

The nanotechnologies themselves are not the only things being focused on and improved upon for the future, but so are the means in which these technologies are utilized. For one, nanotechnologies have future directions in the area of point-of-care diagnostics, allowing people to utilize nanotechnologies to address medical issues in home settings. There are devices being developed for POC early diagnostics of neurodegenerative diseases like dementia and Alzheimer's, early diagnosis of cancers, and that of infectious diseases such as Malaria [14]. The concept of detecting serious illnesses like cancer at an early stage is remarkable, given the widely recognized importance of early diagnosis for improving survival chances. Not only is

nanotechnology being integrated into POC diagnostics, but it is also being integrated into the Internet of Medical Things (IoMT). The goal of the IoMT is to leverage interconnected medical devices and healthcare systems to enable remote monitoring, analysis of data, and personalized healthcare. With the introduction of the Internet of Nano Things (IoNT), its future will prove to be "instrumental in providing flexible and pervasive healthcare[...] smart drug administration and nanoscale surgeries"[15]. This progress is so significant because in today's world, the internet is utilized by everybody daily. To be able to integrate biomedical semiconductor data and processes seamlessly with all other medical information on the internet, doctors can efficiently access comprehensive patient data, make informed decisions, and deliver personalized care with greater ease and success.

Finally, in regards to future directions of biomedical semiconductor devices, the industry is moving towards two things: organic semiconductor materials and artificial intelligence. As discussed earlier, the issues with semiconductor materials over the likes of materials such as hydrogels were their lack of flexibility and conformability for the use of wearable and implantable devices. What makes organic semiconductor materials a great avenue for future device development is their ability to bend and stretch without losing their functionality, along with their high biocompatibility and low-cost, due to often being synthesized from abundant carbon-based materials [16]. In recent years, AI and machine learning has become ubiquitous across all aspects of society, and this remains true for the biomedical semiconductor industry. What makes the use of AI potentially so useful for biomedical devices, especially that of the likes of biosensors and implantable devices, is that these devices will be able to become self-sufficient in regards to their functionality, continuously updating its own operations in order to perform more accurately, sensitively, and reliably. As seen in Figure 10, with the combination of technologies such as biosensors with machine learning, these biosensors transform into "intelligent biosensors", essentially having a mind of their own and the capability of advancing their own technology. Now, imagine implantable technologies that are capable of self-sufficient energy through processes such as energy harvesting and also capable of self-sufficient learning through AI development… the wonders that technologies such as these would be able to perform for the quality of millions of lives would be unparalleled to anything society has already experienced. In this way, the future directions of biomedical semiconductor devices are not only promising, they are poised to revolutionize the healthcare industry as humanity knows it.

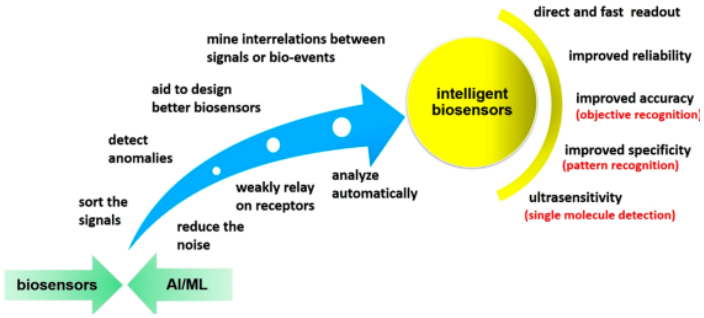

Figure 10: Benefits of machine learning brought to biosensors. [17]

Conclusions:

Generally speaking, there is one major goal of technological development in the modern world: to improve the quality of life and society. While some technologies may contribute more to this goal than others, biomedical devices stand paramount in their advances in life quality, health, safety, and longevity. As the saying goes, a sick man only wishes for one thing: health. “When health is absent, wisdom cannot reveal itself, art cannot manifest, strength cannot fight, wealth becomes useless, and intelligence cannot be applied”. It is for this reason that biomedical semiconductor devices are the most valuable technological resource for humanity, both now and for the future. Imagine telling somebody 100 years ago, that if your heart was failing, you could get a device inserted into your heart to regain its function? Or having implants into your retinas to regain your sight? What were only fantasies and imagination of human history, are now the realities of the modern world. This not only shows how far humanity has come, but also foreshadows how much more is on the horizon for society. Solutions to profound societal burdens such as cancer, diabetes, and heart disease are only getting more and more advanced, and potentially one day in the near future, engineers will be able to utilize the wondrous abilities of semiconductors to eliminate these issues once and for all. While there are many challenges and limitations that engineers face when developing biomedical devices, there are just as many, if not more, solutions being developed. The unimaginable biomedical semiconductor devices that are able to be used today for imaging, sensing, and therapeutic purposes only emphasize just how much engineers have already achieved in combating biomedical challenges and limitations. All in all, the biomedical semiconductor industry is one that holds a breathtaking past and an extremely promising future, and with the next generation of engineers on the horizon, the industry is destined for nothing short of greatness.

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